

September 2026

Learning L^AT_EX language

Naomi Schickele

Learning L^AT_EX language

1. What is LaTeX and why use it?

2. Your first document

3. Structure

4. Figure, equation etc.

5. Bibliography

6. Hands-on challenge

1) What is L^AT_EX and why use it?

LaTeX (pronounced either “LAY-tek” or “LAH-tek” depending on who you talk to) is a **document preparation system** used to create professional-quality documents, such as reports, books, **articles**, **theses**, **CV**, **posters** and others.



Format the document manually

Figures and tables may move

Update references manually

Manage bibliography manually

Formatting can become inconsistent

vs



Describe the structure & content

Automatic positioning & numbering

Automatic cross-references

Automatic bibliography & citations

Consistent formatting



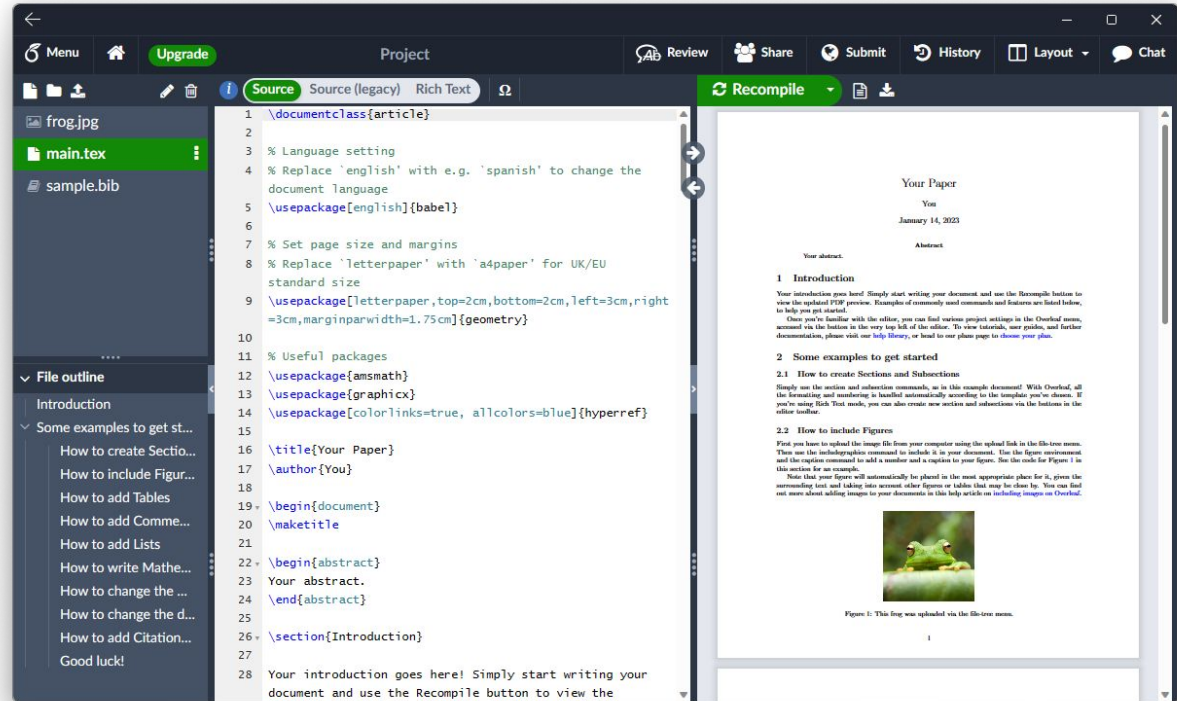
From Wikipedia, the free encyclopedia

Overleaf is software for running a collaborative cloud-based LaTeX editor used for writing, editing and publishing scientific documents.

Unlike **Microsoft Word** or **LibreOffice Writer**, where you see the final layout while typing, LaTeX uses a **plain text file** containing both **your content and formatting commands**.

This file is then processed by a **TeX engine**, which **automatically generates a well-formatted PDF document**.

With LaTeX, you focus on writing the **content**, while LaTeX takes care of the **layout and formatting**.

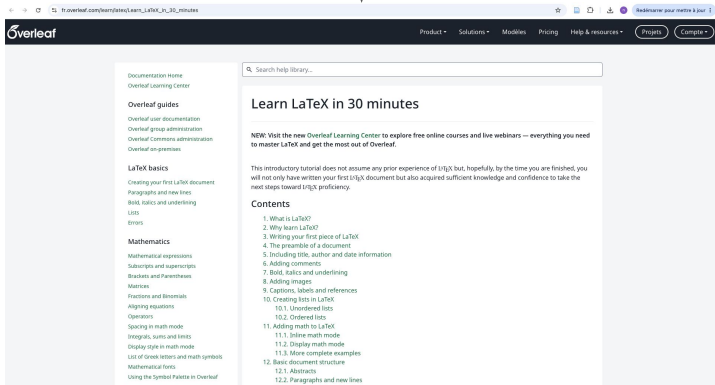




<https://fr.overleaf.com/project>



Welcome in Overleaf



https://www.overleaf.com/learn/latex/Learn_LaTeX_in_30_minutes



=



Mathrice is a CNRS thematic network bringing together IT specialists from mathematical research laboratories, under the **CNRS Institute of Mathematical Sciences and Interactions (INSMI)**.

PLMLatex is based on the **free, open-source Community Edition of Overleaf**.

Compared with the **paid version of Overleaf**, the following features are **not available**:

- Dropbox integration
- Git / GitHub integration
- Track Changes (while shared)

2) Your first document

Home page

Help and guides

PLMlatex mathrice CNRS

Nouveau projet

Tous les projets

- Mes projets
- Partagé avec moi
- Projets archivés
- Corbeille des projets
- ÉTIQUETTES/DOSSIERS
- + Nouveau dossier

Rechercher un projet...

☐ Titre	Propriétaire	Dernière modification ▾	Actions
☐ CV_SCHICKELE_Naomi	You	7 days ago par You	
☐ Scientific Summary	You	9 days ago par You	
☐ Thesis	You	2 months ago par You	
☐ Poster	You	3 months ago par You	

Help ▾

- Documentation PLM
- Documentation Overleaf
- Forum Mathrice

2) Your first document

The screenshot displays the PLMlatex web interface. The top navigation bar includes a 'Menu' icon, the title 'Example', and buttons for 'Partager', 'Historique', and 'Discuter'. Below this, a toolbar contains icons for file operations and a green 'Recompiler' button. The left sidebar shows a file explorer with 'main.tex' selected. The main editor area contains the following LaTeX code:

```
1 \documentclass{article}
2
3 \usepackage[utf8]{inputenc}
4
5 \title{Example}
6 \author{Naomi SCHICKELE}
7 \date{September 2026}
8
9 \begin{document}
10
11 \maketitle
12
13 \section{Introduction}
14
15 Hello, my name is \textit{Naomi Schickele}. Welcome and enjoy learning
16 \textbf{LaTeX}.
17
18 \end{document}
19
```

The right pane shows the rendered PDF output. It features a centered title 'Example', author 'Naomi SCHICKELE', and date 'September 2026'. Below this is a section header '1 Introduction' followed by the text 'Hello, my name is *Naomi Schickele*. Welcome and enjoy learning **LaTeX**.'

2) Your first document

The screenshot displays the PLMlatex editor interface. The main window shows a LaTeX document with the following content:

```
1 \documentclass{article}
2
3 \usepackage[utf8]{inputenc}
4
5 \title{Example}
6 \author{Naomi SCHICKELE}
7 \date{September 2026}
8
9 \begin{document}
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11 \maketitle
12
13 \section{Introduction}
14
15 Hello, my name is \textit{Naomi Schickele}. Welcome and enjoy learning
16 \textbf{LaTeX}.
17
18 \end{document}
19
```

On the left, a sidebar shows the file explorer with a red box highlighting the 'main.tex' file. On the right, a settings panel is open, also with a red box. The settings panel includes sections for 'Télécharger' (Download), 'Actions' (Copy project, Number of words), 'Réglages' (Compiler: pdfLaTeX, Document principal: main.tex, Correcteur orthographique: French, Auto-complétion: On, Auto-fermeture des accolades: On, Vérification du code: On, Apparence de l'éditeur: textmate, Raccourcis clavier: None), and 'Aide' (Montrer les raccourcis clavier). A red arrow points from the 'Menu' button in the top-left corner to the 'Télécharger' section of the settings panel.

In the top right corner, the logos for PLMlatex, mathrice, and CNRS are visible.

2) Your first document

Everything that comes before `\begin{document}` is used to **configure** the document.

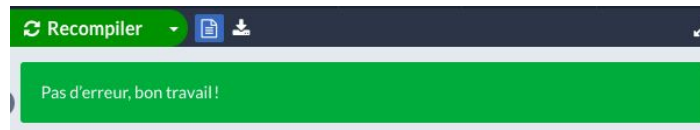
Everything between `\begin{document}` and `\end{document}` **constitutes** the content.

The screenshot displays a LaTeX editor interface with a file explorer on the left, a central code editor, and a preview window on the right. The file explorer shows a file named 'main.tex'. The code editor, titled 'Example', contains the following LaTeX code:

```
1 \documentclass{article}
2
3 \usepackage[utf8]{inputenc}
4
5 \title{Example}
6 \author{Naomi SCHICKELE}
7 \date{September 2026}
8
9 \begin{document}
10
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12
13 \section{Introduction}
14
15 Hello, my name is \textit{Naomi Schickele}. Welcome and enjoy learning
16 \textbf{LaTeX}.
17
18 \end{document}
19
```

A green box labeled 'PREAMBLE packages settings' highlights lines 1 through 8. A blue box labeled 'CONTENT texte figures' highlights lines 9 through 18. The preview window on the right shows the rendered output, including the title 'Example', author 'Naomi SCHICKELE', date 'September 2026', and the section '1 Introduction' with the text 'Hello, my name is *Naomi Schickele*. Welcome and enjoy learning **LaTeX**.'

2) Your first document



Download the pdf

The screenshot shows the PLMlatex web interface. On the left, a file explorer shows "main.tex". The main editor displays LaTeX source code for a document titled "Example" by Naomi SCHICKELE, dated September 2026. The code includes document class, package, title, author, date, and content sections. On the right, a preview window shows the compiled PDF. The PDF has a title page with "Example", "Naomi SCHICKELE", and "September 2026". It also has a section titled "1 Introduction" with the text "Hello, my name is Naomi Schickele. Welcome and enjoy learning LaTeX." The interface includes a top menu bar with "Menu", "Recompiler", and icons for document and download. A red box highlights the top right area of the preview window, which contains buttons for "Partager", "Historique", and "Discuter".

2) Your first document

The screenshot displays the PLMlatex web interface. The top navigation bar includes a 'Menu' icon, the document title 'Example', and buttons for 'Partager', 'Historique', and 'Discuter'. Below this is a toolbar with icons for file operations and a green 'Recompiler' button. The left sidebar shows a file explorer with 'main.tex' selected. The main editor area contains LaTeX source code for a document class, package, title, author, date, and content. A red square highlights a vertical scrollbar with up and down arrows. The right sidebar shows a preview of the rendered document, which includes the title 'Example', author 'Naomi SCHICKELE', date 'September 2026', and a section titled '1 Introduction' with a paragraph of text.

```
1 \documentclass{article}
2
3 \usepackage[utf8]{inputenc}
4
5 \title{Example}
6 \author{Naomi SCHICKELE}
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16 \textbf{LaTeX}.
17
18 \end{document}
19
```

Example

Naomi SCHICKELE

September 2026

1 Introduction

Hello, my name is *Naomi Schickele*. Welcome and enjoy learning **LaTeX**.

3) Structure

`\documentclass{`

`article` → paper
`report` → thesis / report
`book` → book
`beamer` → presentation

`\begin{}`

...

`\end{}`

`{ }` = required field

`[ ]` = optional field

```
1 \documentclass[11pt,a4paper]{article}
2
3 \usepackage{graphicx}
4 \usepackage[backend=biber, style=authoryear]{biblatex}
5 \usepackage{hyperref}
6
7 \addbibresource{bib1.bib}
8
9 \title{Learning LaTeX}
10 \author{Your Name}
11 \date{\today}
12
13 \begin{document}
14
15 \maketitle
16
17 \tableofcontents
18
19 \section{Introduction}
20
21 LaTeX is a powerful tool for writing scientific documents and you can check
22 this in the following article \cite{pmlatex_better_than_word}.
23
24 \section{PLMlatex}
25
26 \subsection{Logo of PLMlatex}
27 \begin{figure}[h]
28 \centering
29 \includegraphics[width=0.8\textwidth]{logoPLM.png}
30 \caption{PLMlatex figure.}
31 \label{fig:logoPLM}
32 \end{figure}
33 As shown in Figure-\ref{fig:logoPLM}, this is my first
34 LaTeX figure.
```

Learning LaTeX

Your Name

September 9, 2026

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2.2	Why is important to learn latex?	2
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1 Introduction

LaTeX is a powerful tool for writing scientific documents and you can check this in the following article Surname 2026.

2 PLMlatex

2.1 Logo of PLMlatex



Figure 1: PLMlatex figure.

As shown in Figure 1, this is my first LaTeX figure.

3) Structure

`\usepackage{}`

`\addbibresource{}`

`\title{}`

`\author{}`

`\date{}` ...

⇒ before `\begin{document}`

```
1 \documentclass[11pt,a4paper]{article}
2
3 \usepackage{graphicx}
4 \usepackage[backend=biber, style=authoryear]{biblatex}
5 \usepackage{hyperref}
6
7 \addbibresource{bib1.bib}
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9 \title{Learning LaTeX}
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Learning LaTeX

Your Name

September 9, 2026

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1 Introduction

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2 PLMlatex

2.1 Logo of PLMlatex



Figure 1: PLMlatex figure.

As shown in Figure 1, this is my first LaTeX figure.

3) Structure

⇒ after `\begin{document}`

TEXTE
FIGURE
EQUATION
BIBLIOGRAPHY...

```
1 \documentclass[11pt,a4paper]{article}
2
3 \usepackage{graphicx}
4 \usepackage[backend=biber, style=authoryear]{biblatex}
5 \usepackage{hyperref}
6
7 \addbibresource{bib1.bib}
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1 Introduction

LaTeX is a powerful tool for writing scientific documents and you can check this in the following article Surname 2026.

2 PLMlatex

2.1 Logo of PLMlatex



Figure 1: PLMlatex figure.

As shown in Figure 1, this is my first LaTeX figure.

3) Structure

```
\documentclass[11pt,a4paper]{article}:
```

```
\section{}
```

```
\subsection{}
```

```
\subsubsection{}
```

```
36 ▾ \subsection{Why is important to learn latex?}
37
38 ▾ \subsubsection{To improve skills in coding}
39
40 Coding is very nice !
41
42 ▾ \subsubsection{Put it in you CV !}
43
44 Who wouldn't want to add a skill to their CV ?
45
46 ▾ \section{Conclusion}
47
48 I can now create a structured scientific document
49 with figures and references \parencite{plmlatex_better_than_word}.
50
51 \printbibliography
52
53 \end{document}
```

2.2 Why is important to learn latex?

2.2.1 To improve skills in coding

Coding is very nice !

2.2.2 Put it in you CV !

Who wouldn't want to add a skill to their CV ?

3 Conclusion

I can now create a structured scientific document with figures and references (Surname 2026).

References

Surname, Name (2026). "Great paper on plmlatex, don't hesitate to read it, really !!!" In: *great journal*.

3) Structure

\documentclass[11pt,a4paper]{report}

\chapter{}

\section{}

\subsection{}

\subsubsection{}

Menu

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biblio_these.bib

main.tex

Structure du fichier

Résumé

Abstract

Liste des abréviations

Introduction

Structure et régulation...

De l'ADN à la chr...

L'ADN, suppo...

La chromatine...

Dynamique e...

Transcription et...

Le gène, l'uni...

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Thesis

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Discuter

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Chapitre 1

Introduction

1.1 Structure et régulation du génome eucaryote

Chez les cellules eucaryotes, l'ADN nucléaire est organisé sous forme de chromosomes grâce à différents niveaux de compaction (Figure 1). À une échelle plus fine, l'ADN est associé à des protéines pour former la chromatine. Ces protéines sont appelées des histones et vont permettre l'enroulement et la compaction de l'ADN. L'unité de la chromatine est le nucléosome, formé d'un octamère d'histones et d'un fragment d'ADN. Comprendre les différents états de compaction de l'ADN a permis de mettre en évidence leur rôle dans de nombreux processus essentiels à la survie cellulaire comme la transcription des gènes.

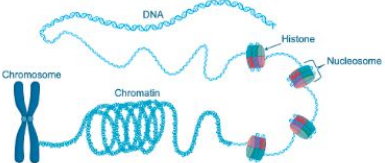


FIGURE 1 – Les différents niveaux de compaction de l'ADN. Du chromosome à l'ADN en passant par la chromatine, le nucléosome et les histones.

1.1.1 De l'ADN à la chromatine

L'ADN, support de l'information génétique

L'ensemble de l'information génétique d'un organisme, également appelé génome, est portée par la molécule d'ADN (Acide Désoxyribonucléique). Cette molécule est composée de deux brins complémentaires, orientés dans deux sens opposés : le brin sens (5' vers 3') et le brin anti-sens (3' vers 5') (Watson, Molecular Biology of the Gene, 2003). L'unité de base de l'ADN est le nucléotide. Il existe quatre types de nucléotides définis par leur base azotée : l'adénine (A), la thymine (T), la cytosine (C) et la guanine (G). Les nucléotides d'un même brin sont reliés entre eux par des liaisons covalentes, tandis que les deux brins d'ADN s'associent grâce à des liaisons hydrogène entre bases complémentaires ou paires de bases (pb) :

13

3) Structure

```
\section{Introduction}
```

```
\underline{\textbf{\textit{LaTeX}}} is a \underline{powerful tool} for  
\textbf{writing scientific documents} and \textcolor{green}{you can check  
this} in \textit{the following article} \cite{pmlatex_better_than_word}.
```

1 Introduction

LaTeX is a powerful tool for writing scientific documents and you can check this in the following article Surname 2026.

- **Bold:** bold text in LaTeX is typeset using the `\textbf{...}` command.
- *Italics:* italicised text is produced using the `\textit{...}` command.
- Underline: to underline text use the `\underline{...}` command.
- **Color:** to color text use the `\textcolor{...}{...}` command.

3) Structure

Latex default front type: **Abstract**

```
4 \usepackage{fontspec}
5 \usepackage{unicode-math}
6 \setmainfont{Libertinus Serif}
7 \setmathfont{Libertinus Math}
```

Abstract

Topologically associating domains (TADs) are regions of the genome in which genomic sequences interact more frequently with each other than with regions located in neighboring domains. This organization contributes to the three-dimensional folding of the genome within the nucleus. At TAD boundaries, proteins known as insulators are often enriched. These proteins are thought to contribute to the separation of adjacent domains and to limit interactions between elements belonging to different TADs. Gene regulation relies on regulatory elements, in particular enhancers, which activate specific target genes. However, enhancers do not activate all genes, indicating that enhancer-promoter interactions must be tightly controlled. A classical model proposes that TAD borders act as relatively robust barriers that preserve gene regulatory specificity and insulator proteins can prevent an enhancer from activating a distant gene when positioned between the two elements. However, increasing evidence suggests that 3D genome organization is dynamic and probabilistic, and that long-range interactions can occur across these boundaries and insulator proteins sites. In addition, systematically testing this mechanism remains difficult due to the sparsity of Hi-C data, which limits the detection of rare or transient long-range chromatin interactions. This limitation prevents a detailed study of how insulator proteins regulate

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3) Structure

New front type → change the compiler

```
1 \documentclass[hidelinks,11pt,a4paper]{report}
2
3 \usepackage[french]{babel}
4 \usepackage{fontspec}
5 \usepackage{unicode-math}
6 \setmainfont{Libertinus Serif}
7 \setmathfont{Libertinus Math}
8 \usepackage{chngcntr}
9 \usepackage{xcolor}
10 \usepackage{titlesec}
11
12 \usepackage{hyperref}
13 \usepackage[all]{hypcap}
14
15 \usepackage{pdfpages}
16
17 \counterwithout{figure}{chapter}
18
19 % bibliography
20 \usepackage[
21   backend=biber,
22   style=authoryear
23 ]{biblatex}
24
25 \addbibresource{biblio_these.bib}
26
27 % nom des sections en haut de pages :
28 \usepackage{fancyhdr}
29
30 \pagestyle{fancy}
31 \fancyhf{}
32
33 \setlength{\headheight}{14pt}
34 \setlength{\headsep}{20pt}
35
```

Erreur de compilation. Désolé, impossible de compiler votre code LaTeX. Veuillez vérifier les erreurs ci-dessous ou consulter les journaux bruts pour plus de détails..

Fatal /usr/local/texlive/2025/texmf-dist/tex/latex/fontspec/fontspec.sty, line 101
Package fontspec Error: The fontspec package requires either XeTeX or

(fontspec) LuaTeX.

(fontspec)

(fontspec) You must change your typesetting engine to,
(fontspec) e.g., "xelatex" or "lualatex" instead of
(fontspec) "latex" or "pdflatex".

Type <return> to continue.

...

L.101 \msg_fatal:nn {fontspec} {cannot-use-pdftex}

/usr/local/texlive/2025/texmf-dist/tex/latex/fontspec/fontspec.sty, line 101
Emergency stop.

<read *>

L.101 \msg_fatal:nn {fontspec} {cannot-use-pdftex}

*** (cannot \read from terminal in nonstop modes)

==> Fatal error /usr/local/texlive/2025/texmf-dist/tex/latex/fontspec/fontspec.sty
occurred, no output PDF file produced!

Télécharger



Code source



PDF

Actions

Copier le projet

Nombre de mots

Réglages

Compilateur

✓ pdfLaTeX
LaTeX
XeLaTeX
LuaLaTeX

Document principal

Correcteur orthographique

French

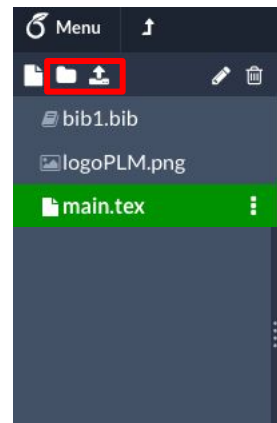
Auto-complétion

On

Auto-fermeture des accolades

On

4) Figure, equation, pdf, scheme and list



```
28 \section{PLMlatex}
29
30 \subsection{Logo of PLMlatex}
31
32 \begin{figure}[h]
33   \centering
34   \includegraphics[width=0.8\textwidth]{logoPLM.png}
35   \caption{PLMlatex figure.}
36   \label{fig:logoPLM}
37 \end{figure}
38
39 As shown in Figure-\ref{fig:logoPLM}, this is my first
40 LaTeX figure.
```

size

Equation

```
52 \section{Equation}
53
54 classique :
55 \begin{equation}
56 E = mc^2
57 \end{equation}
58
59 or :
60 \[
61 E = mc^2
62 \]
63
64 and also :
65 \begin{equation}
66 P = \frac{TP}{TP + FP}
67 \end{equation}
```

Figure

2 PLMlatex

2.1 Logo of PLMlatex



Figure 1: PLMlatex figure.

As shown in Figure 1, this is my first LaTeX figure.

3 Equation

classique :

$$E = mc^2 \quad (1)$$

or :

$$E = mc^2$$

and also :

$$P = \frac{TP}{TP + FP} \quad (2)$$

4) Figure, equation, pdf, scheme and list

```
10 \usepackage{pdfpages}
```

The screenshot shows the Overleaf LaTeX editor interface. On the left, a sidebar displays the file structure of the project, including `bib1.bib`, `instructions.pdf` (highlighted with a red box), `logoPLM.png`, and `main.tex` (highlighted with a green box). Below the file list, a section titled "Structure du fichier" shows a tree view of the document's content, including "Introduction", "PLMlatex", "Logo of PLMlatex", "Why is important to...", "To improve skills...", and "Add it on your CV!". The main editor area displays the LaTeX source code for `main.tex`. The code includes a `\usepackage{pdfpages}` line (highlighted with a red box) and a `\includepdf[pages=-]{instructions.pdf}` line (also highlighted with a red box). The code also defines a list of items and a conclusion section. On the right, a sidebar shows the "Learn LaTeX in 30 minutes" tutorial, which includes a search bar, a list of guides, and a table of contents.

Menu ↑ Example

Partager Historique Discuter

Recompiler

Search help library...

Learn LaTeX in 30 minutes

NEW: Visit the new Overleaf Learning Center to explore free online courses and live webinars — everything you need to master LaTeX and get the most out of Overleaf.

This introductory tutorial does not assume any prior experience of $\text{\LaTeX}$ but, hopefully, by the time you are finished, you will not only have written your first $\text{\LaTeX}$ document but also acquired sufficient knowledge and confidence to take the next steps toward $\text{\LaTeX}$ proficiency.

Contents

1. What is LaTeX?
2. Why learn LaTeX?
3. Writing your first piece of LaTeX
4. The preamble of a document
5. Including title, author and date information
6. Adding comments
7. Bold, italics and underlining
8. Adding images
9. Captions, labels and references
10. Creating lists in LaTeX
 - 10.1. Unordered lists
 - 10.2. Ordered lists
11. Adding math to LaTeX
 - 11.1. Inline math mode
 - 11.2. Display math mode
 - 11.3. More complete examples
12. Basic document structure
 - 12.1. Abstracts

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Overleaf guides

Overleaf user documentation Overleaf group administration Overleaf Commons administration Overleaf on-premises

LaTeX basics

Creating your first LaTeX document Paragraphs and new lines Bold, italics and underlining Lists Errors

Mathematic

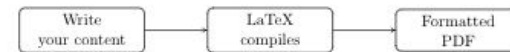
Mathematical expressions Subscripts and superscripts

4) Figure, equation, pdf, scheme and list

Scheme

```
69 \section{Scheme}
70
71 \begin{tikzpicture}[
72   box/.style={
73     rectangle,
74     rounded corners,
75     draw,
76     align=center,
77     minimum width=3cm,
78     minimum height=1cm
79   },
80   >=stealth
81 ]
82
83 \node[box] (write) {Write\\your content};
84 \node[box, right=1.5cm of write] (latex) {LaTeX\\compiles};
85 \node[box, right=1.5cm of latex] (pdf) {Formatted\\PDF};
86
87 \draw[->] (write) -- (latex);
88 \draw[->] (latex) -- (pdf);
89
90 \end{tikzpicture}
```

```
1 \documentclass[11pt,a4paper]{article}
2
3 \usepackage{graphicx}
4 \usepackage[backend=biber, style=authoryear]{biblatex}
5 \usepackage{hyperref}
6
7 \usepackage{tikz}
8 \usetikzlibrary{positioning}
9
```



4) Figure, equation, pdf, scheme and list

```
\clearpage
\chapter*{Liste des abréviations} % ou \section* pour article
\addcontentsline{toc}{chapter}{Liste des abréviations}

\begin{description}
  \item[3C] Chromatin Conformation Capture
  \item[4C] Circular Chromosome Conformation Capture
  \item[5C] Carbone Copy Chromosome Conformation Capture
  \item[ADN] Acide DésoxyriboNucléique
  \item[ADNc] ADN complémentaire
  \item[APA] Aggregate Peak Analysis
\end{description}
```

Liste des abréviations

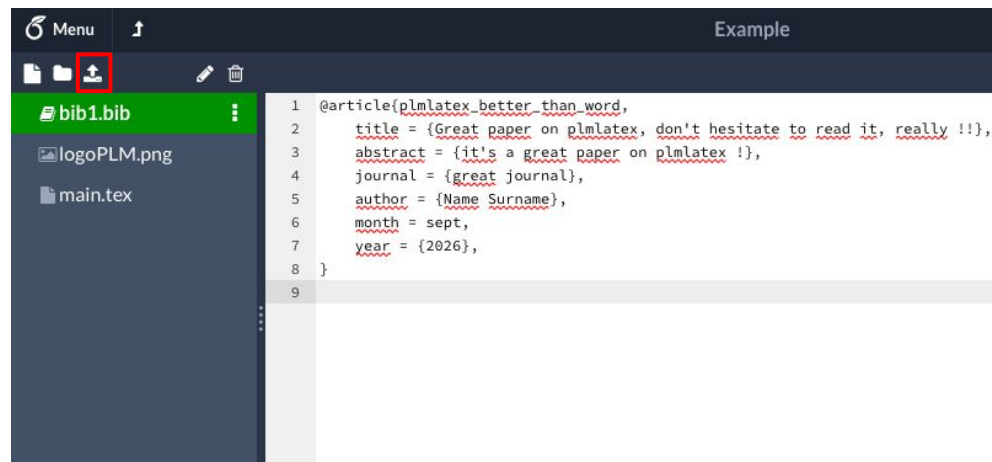
3C Chromatin Conformation Capture
4C Circular Chromosome Conformation Capture
5C Carbone Copy Chromosome Conformation Capture
ADN Acide DésoxyriboNucléique
ADNc ADN complémentaire
APA Aggregate Peak Analysis

```
\clearpage
\addcontentsline{toc}{section}{List of Figures}
\listoffigures
```

List of Figures

1 PLMlatex figure. 3

5) Bibliography



```
1 \documentclass[11pt,a4paper]{article}
2
3 % dimension of the page
4 \usepackage[a4paper,
5   top=2.5cm,
6   bottom=2.5cm,
7   left=3cm,
8   right=3cm]{geometry}
9
10
11 \usepackage{graphicx} % import and manipulate figures
12 \usepackage{hyperref} % make table of contents, figures, references, URLs
   and citations clickable
13
14 \usepackage{tikz} % for the scheme
15 \usetikzlibrary{positioning} % positioning the scheme correctly
16
17 \usepackage[backend=biber, style=authoryear]{biblatex} % format of the
   citation called in the texte
18 \addbibresource{bib1.bib} % add your library .bib from zotero/mendeley or
   other..
```

```
110
111 \section{Conclusion}
112
113 I can now create a structured scientific document
114 with figures and cite references as \cite{plmlatex_better_than_word} and
   \parencite{plmlatex_better_than_word}.
115
116 \printbibliography[
117   heading=bibintoc, % Put it in the table of content
118   title={Bibliography} % Title (default = "References")
119 ]
120
121 \end{document}
```

6 Conclusion

I can now create a structured scientific document with figures and cite references as Surname 2026 and (Surname 2026).

Bibliography

Surname, Name (2026). "Great paper on plmlatex, don't hesitate to read it, really !!" In: *great journal*.

6) Hand-on tutorial

https://github.com/schnaomi/Latex_formation

add a bibliography (export from zotero/mendeley...) + a paper

add a figure

add a section or subsection...

add text (color, bold, italic, underline)

add item section...

The screenshot shows a LaTeX editor interface with a dark theme. On the left, a file explorer shows a project structure with files `bib1.bib`, `logoPLM.png`, and `main.tex`. Below it, a 'Structure du fichier' (File Structure) panel lists the document's sections: Introduction, PLMlatex (with sub-sections 'Logo of PLMlatex' and 'Why is important to learn latex...'), Equation, Scheme, Item list, and Conclusion. The main editor displays the LaTeX source code for `main.tex`, starting with document class, package loading, and title/date information. The right pane shows a preview of the compiled document, titled 'Learning LaTeX', with a table of contents listing sections and their page numbers.

```
1 \documentclass[11pt,a4paper]{article}
2
3 \usepackage{graphicx}
4 \usepackage[backend=biber, style=authoryear]{biblatex}
5 \usepackage{hyperref}
6
7 \usepackage{tikz}
8 \usetikzlibrary{positioning}
9
10 \addbibresource{bib1.bib}
11
12 \title{Learning LaTeX}
13 \author{Your Name}
14 \date{\today}
15
16 \begin{document}
17
18 \maketitle
19
20 \tableofcontents
21
22 \newpage
23
24 \clearpage
25 \addcontentsline{toc}{chapter}{Liste des figures}
26 \listoffigures
27
28 \newpage
29
30
31 \section{Introduction}
32
33 \underline{\textbf{\textit{LaTeX}}} is a \underline{powerful tool} for
34 \textbf{writing scientific documents} and \textcolor{green}{you can check
35 this} in \textit{the following article} \cite{plmlatex_better_than_word}.
36
37 \section{PLMlatex}
38
39 \subsection{Logo of PLMlatex}
```

Learning LaTeX

Your Name
September 10, 2026

Contents

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2 PLMlatex	3
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3 Equation	3
4 Scheme	3
5 Item list	4
6 Conclusion	4
Bibliography	4

1

Conclusion

1. What is LaTeX and why use it?

Word vs LaTeX → content and structure vs formatting

2. Your first document

`\documentclass, document, title`

3. Structure

`\chapter, \section, \subsection, \subsubsection`
`\begin, \end`

4. Figure, equation etc.

`figure, caption, label, equation, pdf, \ref, \tikzpicture`

5. Bibliography

`biblatex, \cite, \parencite, \printbibliography`

6. Hands-on challenge

add a section, a figure, and a reference.

Questions?